



SYNOPSIS FOR EL

Experiential Learning for SDG4
Quality Education

For the V Semester B.E PROGRAMS

(ACY 2025-26)

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Team Name (Unique)	Yogyata
Theme	SDG 4 Quality Education
Team No (Office work)	19
Title of the Project	Yogyata - Decentralising Access to Quality Education with Micro-credentials
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1. Introduction

The global higher education landscape is undergoing a paradigm shift from static, time-based degrees to dynamic, competency-based micro-credentials. As the gig economy and rapid technological advancements reshape workforce requirements, the traditional model of a "one-time" university degree is becoming insufficient. Learners now require a mechanism to verify granular skills such as specific programming languages or project management certifications continuously throughout their careers. However, the current infrastructure for credentialing is fragmented, heavily centralized, and prone to inefficiency.

2. Problem Definition

2.1. Problem Statement:

The Challenge: Despite the digitalization of academic records, the global credentialing ecosystem remains inefficient, insecure, and rigid. The core problem is the **systemic failure of current verification infrastructures to provide trust, privacy, and utility** for learners and employers alike. **Relevance and Importance** In a real-world context, this failure creates significant friction in the labor market. Employers struggle to verify claims efficiently, leading to hiring delays or credential fraud. Simultaneously, learners are disempowered; they lack ownership over their data and are forced into "all-or-nothing" disclosures that compromise their privacy. Furthermore, the static nature of current credentials fails to support the modern necessity for lifelong learning, leaving a disconnect between a student's past achievements and future market demands. Addressing this is critical for creating a transparent, efficient, and learner-centric economy.

2.2. Background Information: (literature review)

Context and Current Understanding The transition from paper to digital formats (such as PDFs) was intended to modernize credentialing. However, current literature and market analysis indicate that these "Web 2.0" solutions have merely digitized existing problems rather than solving them. The current landscape is defined by three persistent limitations: **limitations of Centralized Verification (Trust & Fraud):** Historically, credential verification relies on centralized authorities (universities or third-party clearinghouses). Research indicates that standard digital certificates are still susceptible to forgery. Furthermore, the verification process is often manual, resulting in slow turnaround times and high administrative costs. The reliance on a central point of failure creates bottlenecks that hamper global mobility for students and workers. **The "All-or-Nothing" Disclosure Model (Privacy & Granularity):** Current standards for academic transcripts lack granularity. To prove a specific competency (e.g., a single course in Python), a learner is typically required to share their entire academic history. This violates the principle of **data minimization** and exposes the learner to potential bias regarding unrelated grades or subjects. There is currently no widespread, user-friendly mechanism for "selective disclosure"—the ability to segregate and share only



relevant credentials with recruiters.

Static Data vs. Dynamic Learning (Data Silos): Traditional degrees operate as "snapshots" in time. Once issued, they become static, "dead" records stored in institutional silos. They do not evolve as the learner gains new skills, nor do they interact with labor market data. Current systems lack the intelligence to analyze a learner's existing credentials against market trends to offer gap analysis or recommendations for future study, rendering the data passive rather than actionable.

3. Objectives

3.1. Primary Objectives

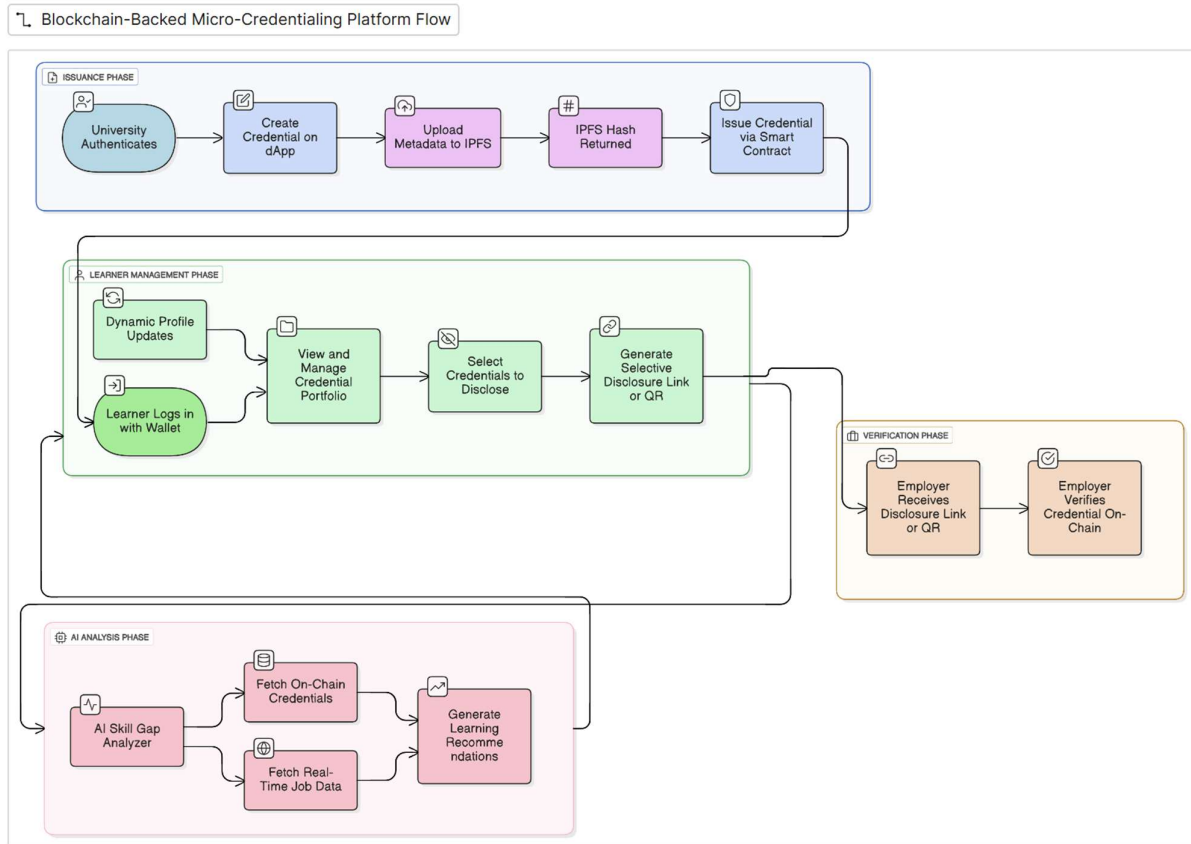
- **Scalable and Trustworthy Verification:**
To implement a tamper proof, gas-efficient credentialing system using Polygon (Layer-2) blockchain and IPFS.
- **Credential Segregation and Privacy Preserving Sharing:**
To enable selective disclosure of specific micro-credentials without sharing the full academic transcript.
- **AI-Powered Skill Gap Analysis and Learning Recommendations:**
Develop an NLP powered engine that matches verified on-chain credentials with real-time job requirements to generate personalized course recommendations and identify missing competencies.

3.2. Secondary Objectives:

- **Create a Dynamic, Actionable Education Profile:**
To evolve traditional static transcripts into **dynamic, intelligence-driven learner profiles** that update continuously as new micro-credentials are added, enabling real-time insights, personalized learning guidance, and alignment with emerging industry skill demands
- **Solve Cost and Scalability Barriers:**
To ensure the platform is **cost-effective and scalable** for widespread adoption

4. Methodology

4.1 Approach:



- Issue micro-credentials on Polygon blockchain:
 - Credentials are created by authorized institutions, stored with metadata on IPFS for security and transparency.
- Learner self-management:
 - Students access and control credentials using their blockchain wallet for privacy and portability.
- Selective sharing:
 - Learners disclose only chosen credentials to employers, protecting privacy by not revealing their entire transcript.
- On-chain verification:
 - Employers verify credential authenticity directly on the blockchain, without intermediaries.
- AI-driven profile analysis:
 - An engine matches learner's credentials to job requirements and recommends missing skills.

Week 1: Planning & Setup

- Finalize project requirements, user stories, and security/privacy principles.
- Prepare platform architecture and flowchart.



- Set up Polygon testnet accounts and baseline tools (Infura, IPFS node, Git repo).

Week 2: Smart Contract & Backend Development

- Write and deploy credentialing smart contracts on Polygon testnet.
- Develop backend endpoints for credential metadata upload to IPFS.
- Build basic event listening service for credential issuance.

Week 3: Frontend & Integration

- Develop React dApp for wallet login, credential viewing, and issuance.
- Integrate MetaMask wallet authentication.
- Connect frontend to backend and test issuance end-to-end.

Week 4: Advanced Features & Testing

- Implement selective disclosure (credential sharing) UI.
- Develop AI-powered skill gap analysis engine (prototype recommendation logic).
- End-to-end platform testing (issuance, sharing, verification).
- Get initial user feedback, fix bugs, and prepare a brief demo or pilot launch.

5. Tools and Techniques Used

5.1 Tools:

- Polygon (Mumbai) Testnet:
Blockchain network used for deploying smart contracts, issuing verifiable credentials, and running pilot dApp transactions.
- Infura:
RPC provider to interact with Polygon nodes for reliable on-chain data access and smart contract deployment.
- IPFS (InterPlanetary File System):
Decentralized storage platform for credential metadata; ensures tamper-proof and distributed access to records.
- MetaMask:
Browser wallet for user authentication, transaction signing, and managing credential ownership.
- Node.js & Express:
Backend API server for handling credential metadata uploads to IPFS and serving analytics.
- React:
Frontend framework for building the user interface, including dashboard, wallet integration, and credential display.
- Ethers.js :



JavaScript library for smart contract interactions, event listening, and blockchain calls from the frontend and backend.

- Python (NLP/AI Libraries – e.g., spaCy, NLTK, scikit-learn, sentence-transformers):
Used for natural language processing, preprocessing, embedding generation, similarity matching, skill gap analysis, and generating lightweight ML-based learning recommendations (e.g., using k-NN or logistic regression).
- Job Posting APIs / Controlled Scrapers (LinkedIn, Indeed, Glassdoor or simple site scrapers):
Sources for labour-market data (job descriptions, skills) used by the recommendation engine. Prefer API where possible; a small scraping script is acceptable for prototypes.
- MongoDB or PostgreSQL (local/simple DB):
Lightweight database for storing user profiles, indexed credential references (on-chain CIDs), and basic logs.
- Git & GitHub:
Version control and collaboration tools for codebase management and tracking changes.

5.2 Techniques

- Smart Contract-Based Credentialing:
Credentials are issued and verified using Solidity smart contracts to maximize trust and reduce credential fraud.
Chosen for transparency and automation of verification.
- Off-Chain Decentralized Metadata Storage:
Credential details are stored on IPFS, separating data from blockchain to reduce costs and improve scalability.
Chosen for cost, privacy, and data integrity.
- Self-Sovereign Identity (SSI):
Users control credentials with their own wallet, supporting privacy-preserving selective disclosure.
Applied to give learners true data ownership and minimize exposure.
- Event-Driven Data Indexing:
Backend listens for on-chain CredentialIssued events to build a user-friendly search index and enable fast querying.
Facilitates real-time updates and efficient off-chain DB queries.
- Selective Disclosure/UI Access Control:
Interface allows learners to choose which credentials to share, implementing the principle of data minimization.
Applied to meet privacy and compliance requirements.
- Simple Hybrid Recommendation (Content-first + Rules):
Combine embedding similarity (content-based) with lightweight rule-based boosts (e.g., prefer courses with matching canonical skills) and simple weighting by user preference. Use scikit-learn k-NN or a small ranking function to produce the final list.
- NLP-Based Skill Gap Analysis:



Python AI modules compare learner's credentials to job postings, then recommend personalized education paths. Chosen to provide actionable insights and support lifelong learning.

- **Agile Development & Iterative Prototyping:**

Rapid development cycles with regular testing and feedback to continuously refine user experience and system logic. Ensures alignment with stakeholder needs and high-quality delivery.

6. Proposed Results

- **End-to-End Credential Platform:**

Functional dApp enabling institutions to issue, store, and verify micro-credentials using Polygon and IPFS.

- **Tamper-Proof Verifiable Credentials:**

Each issued credential is cryptographically secure, easy to verify on-chain, and resistant to forgery or alteration.

- **Learner-Controlled Data:**

Users can log in with their wallet, see their full credential history, and selectively share credentials, ensuring privacy and data ownership.

- **Instant, On-Chain Verification:**

Employers and third parties can instantly verify the authenticity of any credential without contacting the issuing institution.

- **Dynamic, Adaptive Learner Profiles:**

Profiles automatically update as new micro-credentials are added or skills evolve, giving timely feedback for career progression.

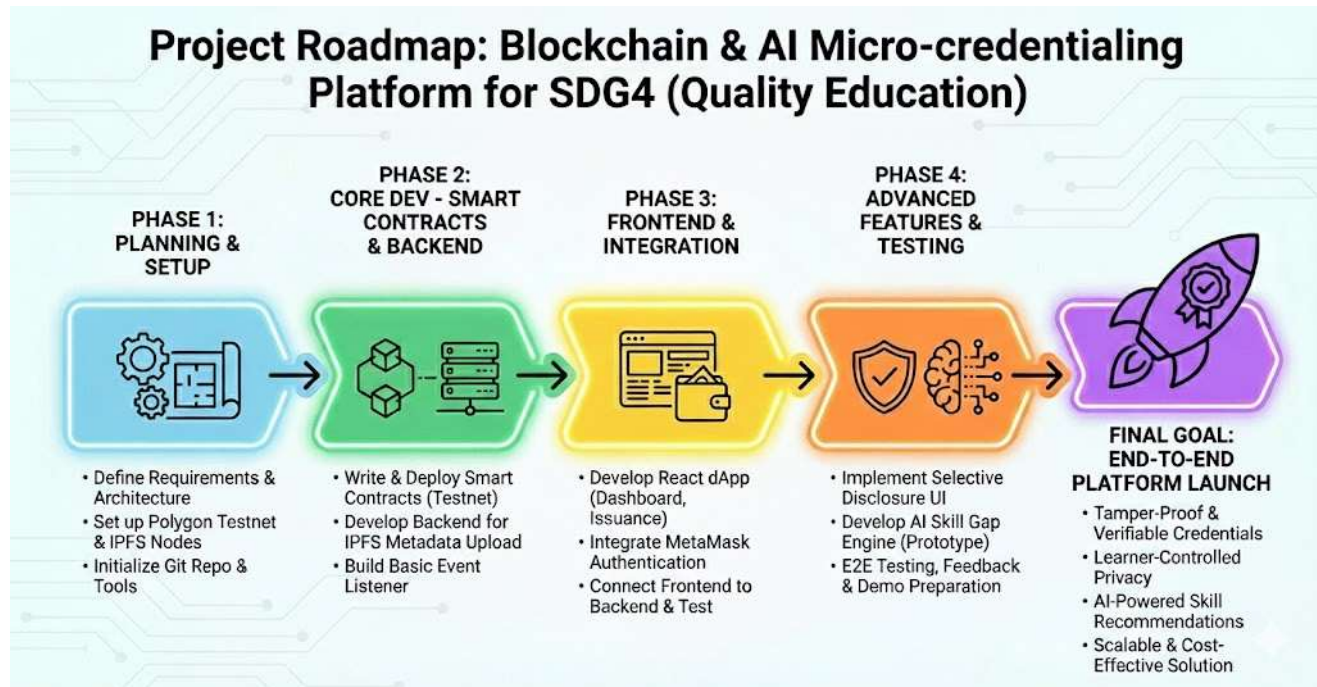
- **AI-Powered Recommendations:**

The system produces personalized skill gap analyses and course suggestions based on verified credentials and real-time job market data.

- **Lower Costs and Improved Scalability:**

The solution demonstrates gas-efficient transactions and scalable metadata storage, suitable for widespread adoption.

7. Road Map



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9. Visuals: The Below Image is the UI/UX design and layout for our app.

